

Multiple Choice

1. **Answer:** C

Options: A) greater than 0.50; B) between 0.16 and 0.50; C) between 0.02 and 0.16; D) between 0.01 and 0.02

Note: Correct option: C - between 0.02 and 0.16

2. **Answer:** D

Options: A) $x^2 + y^2 + z^2 + 8 = 6$; B) $(x^2 + y^2 + z^2)^2 = 8 + 36(x^2 + z^2)$; C) $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + z^2)$; D) $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + y^2)$

Note: Correct option: D - $(x^2 + y^2 + z^2 + 8)^2 = 36(x^2 + y^2)$

3. **Answer:** D

Options: A) True, True; B) True, False; C) False, True; D) False, False

Note: Correct option: D - False, False

4. **Answer:** B

Options: A) -2; B) -1; C) 0; D) 2

Note: Correct option: B - -1

5. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.95'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, True'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '4'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Because the rocking chairs are indistinguishable from each other and the stools are indistinguishable from each other, we can think of first placing the two stools somewhere in the ten slots and then filling the rest with rocking chairs. The first stool has 10 slots in which it can go, and the second has 9. However, because they are indistinguishable from each other, we have overcounted the number of ways to place the stools by a factor of 2, so we divide by 2. Thus, there are $\frac{10 \cdot 9}{2} = 45$ distinct ways to arrange the ten chairs and stools for a meeting.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** True, True. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key